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Editors
The Annals of Applied Statistics

Dear Editors,

Please consider our substantially revised manuscript, “The Blown Lead Paradox: A Pathwise Calibration Benchmark for Win Probability Forecasts,” for publication in *The Annals of Applied Statistics*. This submission follows manuscript AOAS2602-023.

The revision places the sports application at the center of the paper. It begins with a specific NBA collapse, explains why the eventual loser’s maximum win probability is a reverse-conditioned path statistic, and develops only the theory needed to benchmark that quantity. The empirical analysis now includes complete regular-season NFL and NBA inventories for 2018–2024, explicit feed-coverage accounting, and a fixed-clock calibration analysis connected to the sports-forecasting literature, particularly the calibration-surface approach of Yeh, Rice, and Dubin. The NFL and NBA are presented as parallel applications rather than as a controlled comparison between sports.

We have also strengthened the inferential foundation. Games sharing teams are handled with a season-stratified dyadic pigeonhole bootstrap, supported by an exchangeable-array empirical-process argument and a simulation study on the observed schedules. The paper distinguishes this asymptotic justification from finite-sample exactness and reports the expected conservatism of the sparse NFL schedule. Cross-season franchise linkage, calendar-block shocks, probability rounding, feed corrections, missingness, threshold conventions, and leave-one-season-out analyses are reported transparently. The complete rebuild leaves the substantive conclusion intact: the NBA feed has an excess of severe eventual-loser peaks, while the NFL analysis finds no detected departure.

The main article now carries one continuous applied argument: the global pathwise departure, its magnitude at the 0.95 benchmark tail, and the first-crossing mechanism. Proofs, full calibration surfaces, validation, robustness tables, and finite-schedule simulations are in the supplement. All reported values are generated from one consolidated results file and one TeX macro file.

Thank you for considering the revised manuscript.

Sincerely,

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of Pennsylvania